

MANAGEMENT OF ERECTILE DYSFUNCTION

IN PRIMARY CARE

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INTRODUCTION

Erectile dysfunction (ED) is a common condition where men struggle to achieve or maintain an erection, impacting sexual performance. It affects men of all ages, increasing with age. ED has physical (vascular, hormonal, neurological), psychological (anxiety, depression), and lifestyle-related causes. This condition significantly affects sexual and emotional well-being, self-esteem, and relationships.

Epidemiology

ED is a growing concern worldwide, with an estimated 322 million men expected to be affected by 2025. In Singapore, the prevalence of ED is already significant, with approximately half of all males above 30 reporting some degree of erectile dysfunction. This number increases substantially with age, affecting three-quarters of men in their sixties.¹

Risk Factors

- Endocrine – diabetes mellitus, thyroid disorders, hypogonadism, hyperprolactinemia
- Metabolic syndrome – hypertension, hyperlipidaemia, obesity, sedentary lifestyle
- Vascular – peripheral vascular disease
- Substance use – smoking, alcohol, illicit drugs
- Neurological – Stroke, Parkinson's disease, multiple sclerosis, spinal cord injury
- Structural – Peyronie's disease, phimosis, prostate cancer treatment (surgery, radiation, hormone therapy), trauma
- Psychological – stress, marital or relationship issues, guilt, anxiety, depression, history of sexual abuse
- **Drugs:**
 - Anticonvulsants: Phenytoin
 - Antidepressants: Lithium, MAO inhibitors, SSRIs, SNRIs, TCAs
 - Antihistamines: dimenhydrinate, diphenhydramine, hydroxyzine
 - Antihypertensives: Alpha blockers, beta blockers, calcium channel blockers, thiazides, spironolactone
 - Anti-Parkinson agents: Bromocriptine, levodopa, trihexyphenidyl
 - Psychotropics: Chlorpromazine, haloperidol, benzodiazepines
 - Cardiovascular drugs: Digoxin, gemfibrozil
 - Hormonal agents: 5-alpha-reductase inhibitors, androgen receptor blockers, corticosteroids, GnRH analogues

DIAGNOSIS

The history and physical examination have been reported to have a 95% sensitivity but 50% specificity in determining cause of ED, needing further tests to assist in determining its cause.²

History

- Assessment of libido, morning erections, premature ejaculation, duration of symptoms
- Sexual history (5Ps) – **P**artner, **P**ractice, **P**ast history of STI, **P**rotection from STI, **P**revention of **P**regnancy
- Past medical history
- Previous evaluation of cardiovascular system, presence of exertional symptoms, concomitant use of nitrates
- Baseline effort tolerance (ability to climb 2 flights of stairs, or walk 2 bus stops)
- Symptoms of hypogonadism – decreased libido, fatigability, loss of lean muscle mass, mood changes
- Evaluation of specific cause and risk factors for ED

Physical Examination

- Blood pressure, heart rate, body mass index (BMI)
- Abdomen and external genitalia, specifically looking out for structural issues and features of hypogonadism (loss of male hair pattern, gynaecomastia, small testes)
- Assessment of femoral and peripheral pulses
- Targeted neurological evaluation

Erection Hardness Score (EHS)

Score	Severity	Consistency	Remarks
1	Severe	Tofu	Penis is large but not hard
2	Moderate	Peeled Banana	Penis is hard but not hard enough for penetration
3	Suboptimal	Unpeeled Banana	Penis is hard enough for penetration but not completely hard
4	Optimal	Cucumber	Penis is hard and completely rigid

Simplified Index of Erectile Function (IIEF-5) Score

Over the past 6 months:					
1. How do you rate your confidence that you could get and keep an erection?	Very low 1	Low 2	Moderate 3	High 4	Very high 5
2. When you had erections with sexual stimulation, how often were your erections hard enough for penetration?	Almost never/never 1	A few times (much less than half the time) 2	Sometimes (about half the time) 3	Most times (much more than half the time) 4	Almost always/always 5
3. During sexual intercourse, how often were you able to maintain your erection after you had penetrated (entered) your partner?	Almost never/never 1	A few times (much less than half the time) 2	Sometimes (about half the time) 3	Most times (much more than half the time) 4	Almost always/always 5
4. During sexual intercourse, how difficult was it to maintain your erection to completion of intercourse?	Extremely difficult 1	Very difficult 2	Difficult 3	Slightly difficult 4	Not difficult 5
5. When you attempted sexual intercourse, how often was it satisfactory for you?	Almost never/never 1	A few times (much less than half the time) 2	Sometimes (about half the time) 3	Most times (much more than half the time) 4	Almost always/always 5
IIEF-5 scoring: The IIEF-5 score is the sum of the ordinal responses to the 5 items. 22-25: No erectile dysfunction 17-21: Mild erectile dysfunction 12-16: Mild to moderate erectile dysfunction 8-11: Moderate erectile dysfunction 5-7: Severe erectile dysfunction					

The simplified Index of Erectile Function (IIEF-5) score is a widely used, self-administered questionnaire that measures and grades severity of ED.³

Investigations

Baseline investigations are essential to identify underlying medical conditions. The following investigations are recommended to be performed, if not previously done:

- Serum fasting glucose or HbA1c
- Lipid panel
- ECG

Thyroid function tests are indicated for patients exhibiting signs or symptoms suggestive of thyroid disorders.

If hypogonadism is suspected, referral to Urology is advised for morning testosterone level assessment and further management, ensuring comprehensive evaluation and treatment.

MANAGEMENT

Non-Pharmacological Therapy

Effective management of ED involves identifying and addressing the underlying cause, followed by targeted treatment.

Optimising concomitant chronic conditions, such as hypertension, diabetes, and hyperlipidaemia, is crucial to improving ED outcomes.

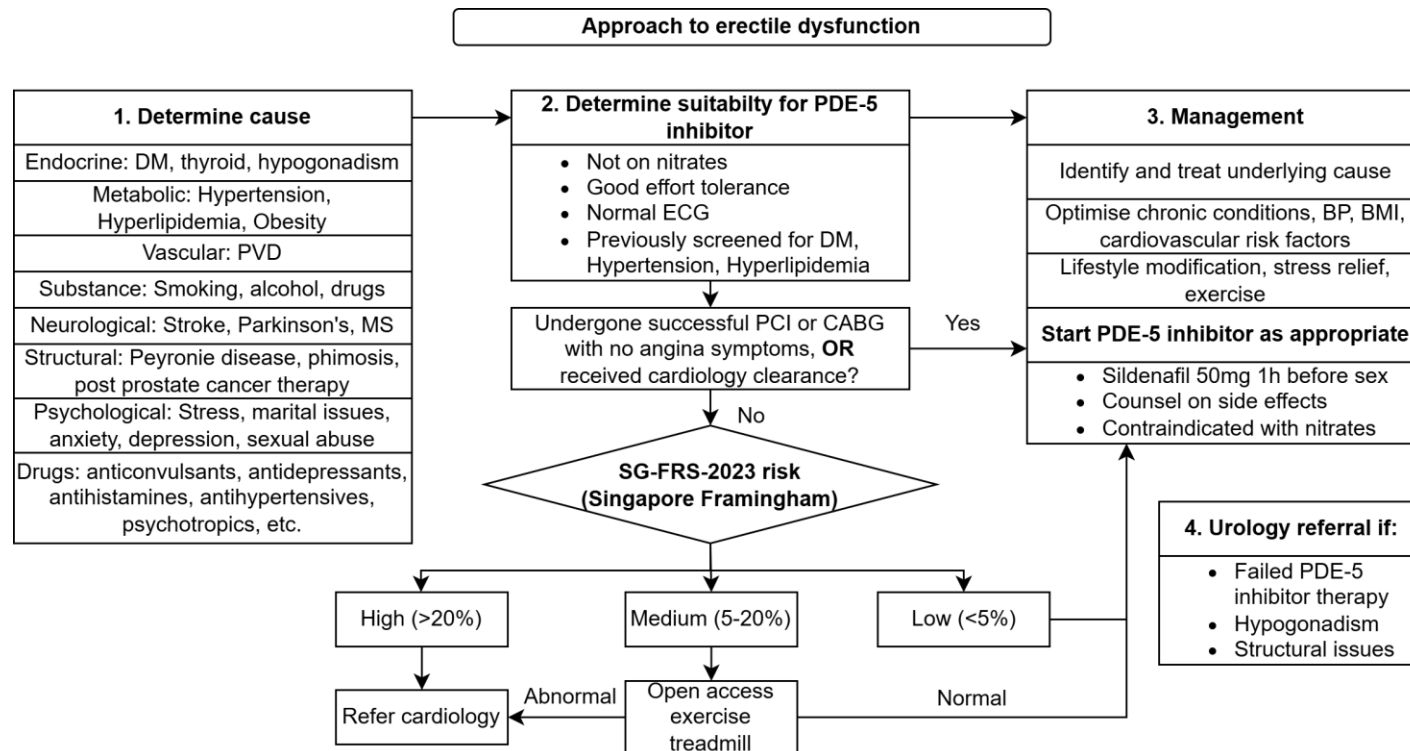
Consider medication adjustment if ED is suspected to be drug-induced.

Lifestyle modifications also play a vital role, with regular exercise, smoking cessation, and weight loss in overweight men demonstrating significant symptom improvement.⁴

Initiating Pharmacological Therapy

ED and cardiovascular disease share common risk factors, increasing the likelihood of cardiovascular events. Family physicians play a crucial role in managing ED through medical evaluation and CVD risk stratification using the Princeton IV consensus recommendations.⁵

Table for diagnosis and management of erectile dysfunction and assessing suitability for PDE-5 inhibitor



- We define cardiovascular risk in accordance with the **Singapore-modified Framingham Risk Score (SG-FRS-2023)**, where low risk is SG-FRS-2023 < 5%, intermediate risk is 5-20%, and high risk is ≥ 20% respectively.
- We recommend performing an ECG to assess for undiagnosed cardiovascular conditions if not performed recently.
- Intermediate risk patients should undergo exercise treadmill investigation. This is to be arranged via the Open Access Exercise Treadmill Testing Workflow which can be found [here](#). Those with normal results are suitable to be started on PDE-5 whereas the rest with abnormal results require a referral to Cardiology for further evaluation.
- High-risk patients should be referred to Cardiology for clearance prior to initiating PDE-5 inhibitor therapy.

PDE-5 Inhibitors

Oral PDE-5 inhibitors form the mainstay and 1st line of medical management of ED in the primary care setting.

Patients without existing cardiovascular diseases can be initiated on PDE-5 inhibitors if they fulfil **ALL** the following:

- Low cardiovascular risk (SG-FRS-2023 < 5%)
- Normal ECG
- Good effort tolerance (able to climb 2 flights of stairs, or walk 2 bus stops)

Patients with existing cardiovascular conditions can be initiated on PDE-5 inhibitors if they:

- Had undergone complete re-vascularisation of coronary arteries i.e. successful PCI / CABG with no angina symptoms, **OR**
- Received cardiology clearance for high-risk patients.

Drug	Initial Dose	Maximum Dose	Common ADRs	Rare ADRs	Remarks / Contraindications / Precautions
Sildenafil (NS) (50 mg tablet) \$	50 mg, 1 hour before sexual intercourse	100 mg in 24 hours, 1 hour before sexual intercourse	Headache, flushing and gastrointestinal symptoms	Visual disturbance, dizziness, myalgia, insomnia, anxiety, vertigo, epistaxis, priapism, hypotension. Can cause potentially fatal cardiovascular events (MI, arrhythmias, unstable angina). Seek medical assistance in cases of prolonged erection (>4h) or sudden visual or hearing loss.	Sildenafil is to be taken on an empty stomach, 2 hours after low fat diet. Manual stimulation prior to intercourse is required. Contraindicated in patients taking nitrates (GTN, ISMN, ISDN) due to risk of severe hypotension, patients with severe cardiovascular disorders (e.g. unstable angina, cardiac failure), loss of vision in 1 eye due to non-arteritic anterior ischemic optic neuropathy, hypotension (BP < 90/50 mmHg), recent stroke or MI, known hereditary degenerative retinal disorders, severe hepatic impairment. Patients on alpha-blockers, mild or moderate liver impairment, CrCl < 30ml/min, should be started on initial dose of 25mg.

Other treatments of ED include Low-intensity Extracorporeal Shockwave Therapy (LiESWT), Intracorporeal Injections, Vacuum Erection devices, and penile prostheses.

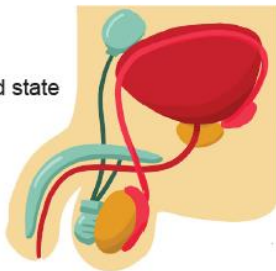
PENILE IMPLANT

阴茎假体植入物

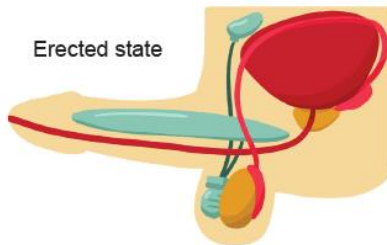
Inflatable penile prosthesis



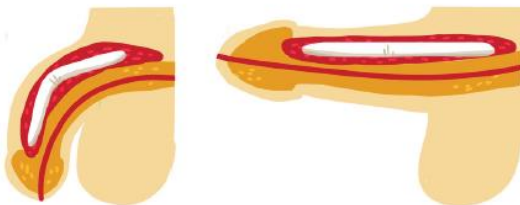
Flaccid state



Erected state



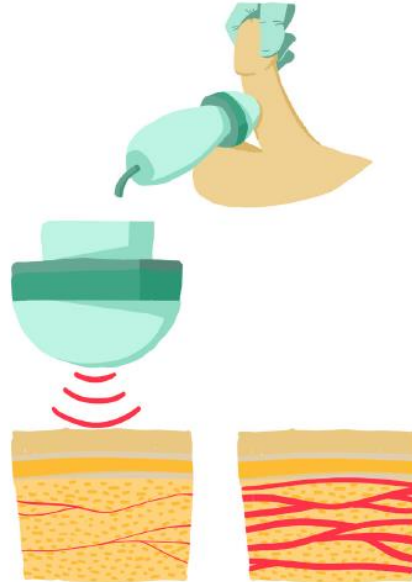
Malleable penile prosthesis



- Penile implants are inserted completely underneath the skin and are not visible
- They can last for many years

LOW-INTENSITY EXTRACORPORAL SHOCKWAVE THERAPY (LIESWT)

阴茎冲击波疗法

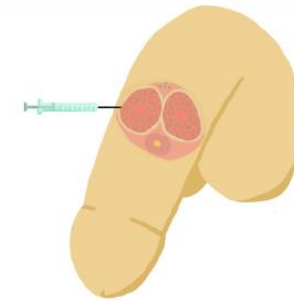


How does it work?

It stimulates growth of new blood vessels in the penis.

INTRACAVERNOSAL INJECTION

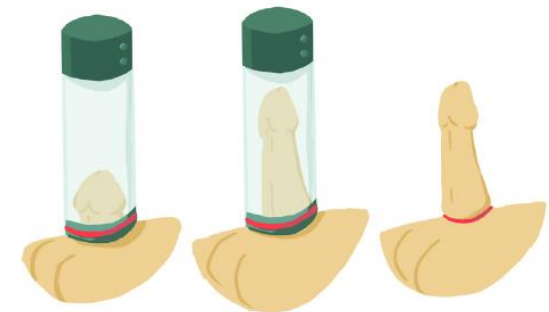
阴茎海绵体内药物注射



How is it administered?
Medication will be injected into the penis 10-30 minutes prior to sexual intercourse.

VACUUM ERECTION DEVICE

真空架设装置



How does it work?

It creates negative pressure to draw blood into the penis, causing an erection.

Referral Criteria to Urology

The following scenarios will warrant a routine referral to Urology for further management of ED:

1. Failed oral PDE5 inhibitor therapy with appropriate use (i.e., after 4 separate occasions of taking maximum doses on an empty stomach and with adequate stimulation)
2. Suspicion of hypogonadism
3. Structural abnormalities, history of pelvic surgery, neurological disorders, complex medical conditions

SPECIAL SITUATIONS

For HSG enrolled or teamlet patients, how should I manage a patient who demands a sildenafil prescription based on previous use from private GP or specialist clinics?

Begin by determining the probable cause of erectile dysfunction and assess cardiovascular risk using the SG-FRS-2023 score. For high-risk patients, politely decline and refer to cardiology. For medium-risk patients, recommend a treadmill stress test to screen for coronary artery disease. If the patient insists, engage in shared decision-making, educating them about potential risks such as cardiac arrhythmias and myocardial ischemia. Ensure the patient understands and accepts these risks, documenting the discussion thoroughly before prescribing. If pressured to prescribe against your judgment, escalate to a senior doctor for assistance. Throughout the process, maintain professionalism, prioritize patient safety, and document all decisions and discussions comprehensively.

For non-HSG enrolled, non-teamlet patients, how do I manage if they are requesting for sildenafil prescription?

Politely decline and advise them to consult their principal doctor (HSG-enrolled GP or specialist) who is responsible for ensuring cardiovascular safety. Inform patients that their principal physician can provide a private prescription fillable at our pharmacy at the same price, without quantity restrictions. If the patient insists, explain that a comprehensive cardiovascular risk assessment, including possible stress tests (medium risk) or cardiology referral (high risk), is required before considering a prescription. If they're unwilling to undergo these assessments, firmly decline and suggest consulting their private GP or specialist. Escalate to senior doctor if unsure, and document communication. The goal is to ensure patient safety while maintaining appropriate care boundaries and respecting established care relationships.

How should I handle requests for sildenafil prescriptions that exceed the monthly limit of 8 tablets (at the time of writing)?

Be aware that the pharmacy will restrict collection to this amount. Explain to the patient that this policy prevents potential abuse, discourages black market sales, rationalizes stock, ensures equal access for all patients, and maintains a healthy drug supply. If the patient requires additional medication, consider asking them to revisit when their supply runs out for a re-prescription.

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